

Gwendal Beaumont

PhD student in Computer Sciences

Brest, France

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Education

- 2023 – **PhD — Computer Sciences, IMT Atlantique**, Brest, France
Title of the thesis: TwinDevOps, an automated process for the deployment and dynamic evolution of Digital Twins.
- 2020 – 2023 **Engineer's Degree — General Engineering, IMT Mines Alès**, Alès, France
One and a half year of general training and one and a half year in the Department of Software Engineering and Artificial Intelligence.
- 2018 – 2020 **Classe préparatoire aux grandes écoles, Lycée Malherbe**, Caen, France
Two-years intensive training to prepare for the French engineering schools competitive exam. Fields of study: Mathematics, Physics, Computer Sciences.

Experience

- 2026-01 – **Doctoral Visiting Researcher, Technische Universität Wien**, Vienna, Austria
2026-04 Doctoral mobility hosted by Assoc. Prof. Bork Dominik where I worked on an extension of a process developed during my PhD thesis.
- 2023-03 – **Intern, ALTEN**, Rennes, France
2023-09 Research & Development internship on the labelling process of data in order to be used with Machine Learning algorithms. Detailed achievements:
 - Creation of a framework using Java Spring Boot to manage unlabelled data and get statistics from it
 - Creation of an interface using TypeScript and the Electron framework
- 2022-05 – **Visiting International Scholar, Miami University**, Oxford, OH, United States of America
2022-08 Internship as an assistant in the preparation of a master thesis on the effect of errors in agent-based simulation systems. Detailed achievements:
 - Creation of a simulation framework in Python to take errors into account
 - Creation of multiple agent-based simulations in Python to test our framework

Services

- 2026 **Seminar co-organisation, IMT Atlantique**, Brest, France
I co-organised one day of the 2026 Annual Seminar of the Department of Computer Sciences. Planning of the seminar is available [here](#).
- 2024 – 2025 **PhD students representative at the Student Board, EULiST Alliance**
Institut Mines-Télécom PhD students representative at the european level in the EULiST Alliance. This alliance integrates over 200,000 students and 20,000 staff at 10 universities in 10 countries.
- 2023 – 2025 **PhD students representative in a workgroup, Institut Mines-Télécom**, France
IMT PhD students representative at the national level in a work group focused on respect and equality.
- 2023 – 2025 **PhD students representative at the board of directors, Institut Mines-Télécom**, France
IMT PhD students representative at the national level for two years. I took part at the board of directors and worked with the scientific director of Institut Mines-Télécom on ways to improve PhD environment.
- 2021 – 2022 **Communication Manager, IMT Mines Alès student council**, Alès, France
Member of the communication managers team for the student council during a year. I administrated the student council's website, managed social networks, and answered and moderated emails and comments on social networks.

Languages

French Native

Spanish Intermediate

Korean Basics

English Fluent (*TOEIC: 975/990*)

German Basics

Book Chapters

- [B1] Gwendal Beaumont, Antoine Beugnard, Jannik Laval, Salvador Martínez, Souad R. Chaniour, Christelle Urtado, Sylvain Vauttier, and Bérenger Akodo. “DevOps for and by Digital Twins Leveraging Virtual Replicas in Continuous Software Engineering”. In: *Digital Twins*. Ed. by Bedir Tekinerdogan and Cor Verdouw. Elsevier B.V., June 2026, pp. 227–242. ISBN: 978-0-443-45573-5. DOI: 10.1016/B978-0-443-45573-5.00018-5.

Journals

- [J2] Jack T. Beerman, Gwendal Beaumont, and Philippe J. Giabbanelli. “A Framework for the Comparison of Errors in Agent-Based Models Using Machine Learning”. In: *Journal of Computational Science* 72 (Sept. 2023), p. 102119. ISSN: 1877-7503. DOI: 10.1016/j.jocs.2023.102119.
- [J3] Jack T. Beerman, Gwendal Beaumont, and Philippe J. Giabbanelli. “A Scoping Review of Three Dimensions for Long-Term COVID-19 Vaccination Models: Hybrid Immunity, Individual Drivers of Vaccinal Choice, and Human Errors”. In: *Vaccines* 10.10 (Oct. 2022), p. 1716. ISSN: 2076-393X. DOI: 10.3390/vaccines10101716.

Conferences

- [C4] Gwendal Beaumont, Antoine Beugnard, Salvador Martínez, Christelle Urtado, and Sylvain Vauttier. “Vers l’automatisation de la gestion du cycle de vie des jumeaux numériques”. In: *Actes de la 44ème édition de la conférence INFORSID*. Toulouse, France, May 2026, pp. 55–57. URL: https://inforsid2026.sciencesconf.org/data/Actes_INFORSID2026.pdf.
- [C5] Gwendal Beaumont, Charlotte Verbruggen, Antoine Beugnard, Dominik Bork, Salvador Martínez, Christelle Urtado, and Sylvain Vauttier. “Engineering Digital Process Twins: Towards a Methodology Based on MDE and DevOps”. In: *ACM/IEEE 29th International Conference on Model Driven Engineering Languages and Systems*. MODELS Companion 2026. Málaga, Spain: Association for Computing Machinery, Oct. 2026. ISBN: 979-8-4007-2903-4. DOI: 10.1145/3837062.3839100.
- [C6] Gwendal Beaumont, Antoine Beugnard, Salvador Martínez, Christelle Urtado, and Sylvain Vauttier. “Towards Automating the Life Cycle Management of Digital Twins”. In: *Conceptual Modeling*. Ed. by Dominik Bork, Roman Lukyanenko, Shazia Sadiq, Ladjel Bellatreche, and Óscar Pastor. Vol. 16189. ER 2025. Poitiers, France: Springer Nature Switzerland, Oct. 2025, pp. 412–430. ISBN: 978-3-032-08623-5. DOI: 10.1007/978-3-032-08623-5_22.
- [C7] Gwendal Beaumont, Antoine Beugnard, Salvador Martínez, Christelle Urtado, and Sylvain Vauttier. “Towards Re-Engineering Digital Twins: Preliminary Experiments on Three Use Cases”. In: *Proceedings of the ACM/IEEE 27th International Conference on Model Driven Engineering Languages and Systems*. MODELS Companion ’24. Linz, Austria: Association for Computing Machinery, Sept. 2024, pp. 453–458. ISBN: 979-8-4007-0622-6. DOI: 10.1145/3652620.3688259.
- [C8] Jack T. Beerman, Gwendal Beaumont, and Philippe J. Giabbanelli. “On the Necessity of Human Decision-Making Errors to Explain Vaccination Rates for Covid-19: An Agent-Based Modeling Study”. In: *2023 Annual Modeling and Simulation Conference (ANNSIM)*. New York, NY, USA: IEEE Computer Society, May 2023, pp. 413–424. ISBN: 978-1-7138-7328-0. URL: <https://ieeexplore.ieee.org/document/10155377>.